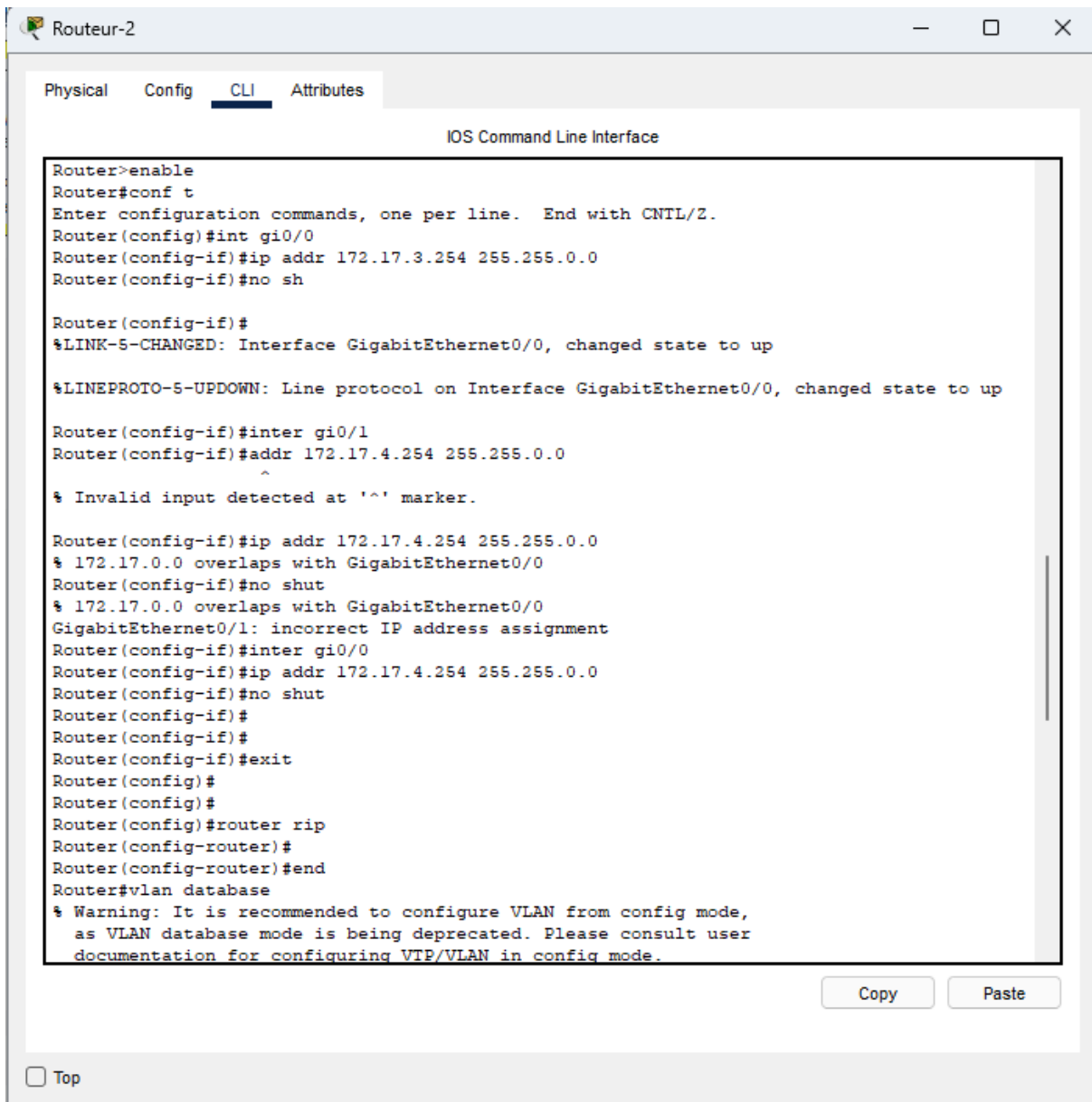


GABIN
Floralyn



Routeur-1

Physical

Config

CLI

Attributes

IOS Command Line Interface

Press RETURN to get started!

Press RETURN to get started!

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gigabitethernet 0/0
^
% Invalid input detected at '^' marker.

Router(config)#interface gigabitethernet
% Incomplete command.
Router(config)#interface gigabitethernet 0/0
Router(config-if)#ip address 172.16.1.254 255.255.0.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
interface gigabitethernet 0/1
Router(config-if)#interface gigabitEthernet 0/1
Router(config-if)#ip address 172.17.2.254 255.255.0.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Copy

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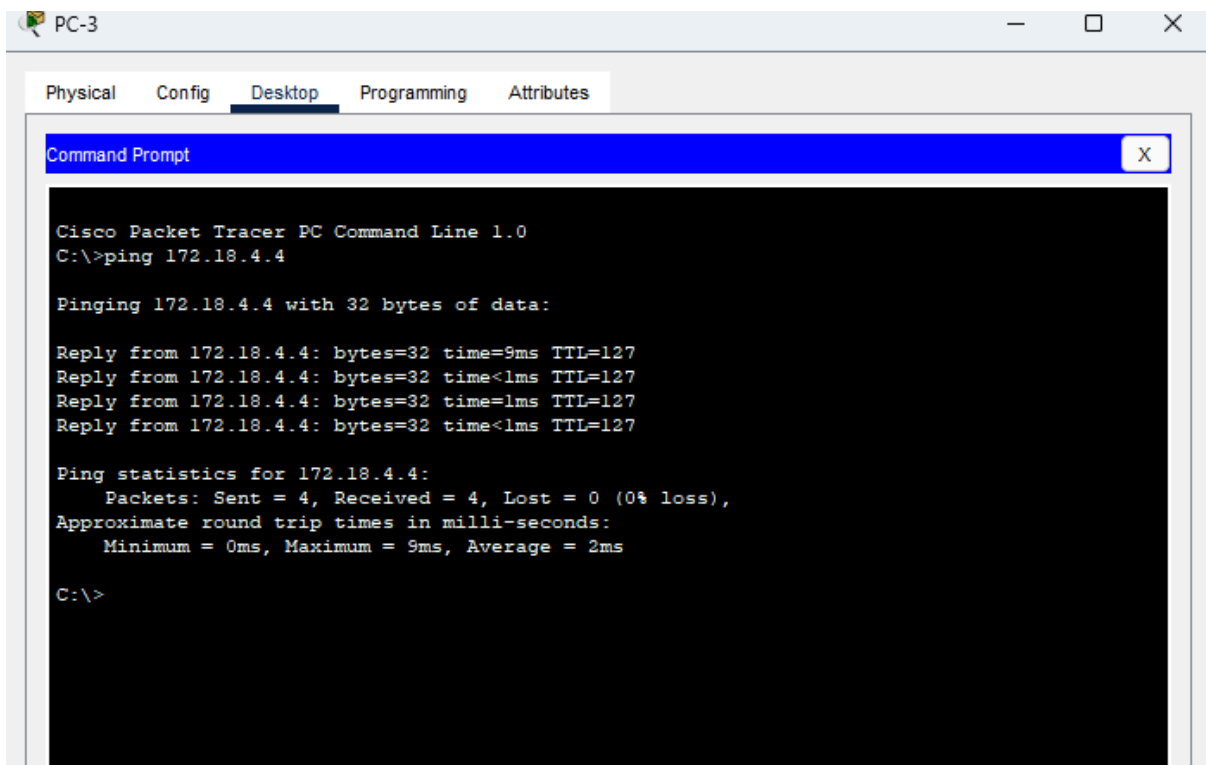
Router(config-if)#
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/1
Router(config-if)#
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/0
Router(config-if)#

Copy

Paste

☐ Top

3



```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 175.18.4.4

Pinging 175.18.4.4 with 32 bytes of data:

Reply from 172.16.1.254: Destination host unreachable.
Reply from 172.16.1.254: Destination host unreachable.
Request timed out.
Reply from 172.16.1.254: Destination host unreachable.

Ping statistics for 175.18.4.4:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 172.18.4.4

Pinging 172.18.4.4 with 32 bytes of data:

Reply from 172.16.1.254: Destination host unreachable.
Reply from 172.16.1.254: Destination host unreachable.
Reply from 172.16.1.254: Destination host unreachable.
Reply from 172.16.1.254: Destination host unreachable.

Ping statistics for 172.18.4.4:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 172.17.2.254

Pinging 172.17.2.254 with 32 bytes of data:

Reply from 172.17.2.254: bytes=32 time<lms TTL=255
Reply from 172.17.2.254: bytes=32 time<lms TTL=255
Reply from 172.17.2.254: bytes=32 time<lms TTL=255
Reply from 172.17.2.254: bytes=32 time<lms TTL=255

Ping statistics for 172.17.2.254:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = lms, Average = 0ms
```

1. Les deux derniers tests fonctionnent car ils sont plus proches des autres réseaux.
2. L'interaction entre le PC-1 et PC-3 n'aboutit pas car selon l'invite de commande, "Request timed out.", Ping statistics for 172.17.3.3", "Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
3. Les deux derniers tests ne fonctionnent pas, s'ils ne peuvent établir une connexion, cela ne permettra pas la communication entre les machines qui sont installées sur différents sites éloignés. D'autant plus que les routeurs n'ont pas les informations nécessaires pour joindre les autres réseaux.
4. PC-1 et PC-3 : Request timed out 4x et Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
PC-2 et PC-4 : Request timed out 4x et Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

PC-1 et PC-4 : Reply from 172.16.1.254: Destination host unreachable 7x. Request timed out. Packets: Sent = 4, Received = 0, Lost = 4 (100% loss).

Je passe à la configuration des route du routeur afin qu'il puisse communiquer avec les machines du routeur 2 et 3.

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Router>enable
Router#show ip interface brief
Interface                IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0       172.16.1.254    YES manual up          up
GigabitEthernet0/1       172.17.2.254    YES manual up          up
Vlan1                    unassigned      YES unset  administratively down down
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    172.16.0.0/16 is variably subnetted, 2 subnets, 2 masks
C       172.16.0.0/16 is directly connected, GigabitEthernet0/0
L       172.16.1.254/32 is directly connected, GigabitEthernet0/0
    172.17.0.0/16 is variably subnetted, 2 subnets, 2 masks
C       172.17.0.0/16 is directly connected, GigabitEthernet0/1
L       172.17.2.254/32 is directly connected, GigabitEthernet0/1

Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip route 172.18.0.0 255.255.0.0 172.17.3.254
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

La table du routeur 2 :

```
S       172.18.0.0/16 [1/0] via 172.17.3.254

Router#
```

Avant de configurer le routeur-2, je teste la communication des machines ci-dessous :
PC-1 & PC-3 : "request timed out"

PC-2 & PC-4 :

"request timed out"

```
C:\>ping 172.18.4.254

Pinging 172.18.4.254 with 32 bytes of data:

Request timed out.
Reply from 172.18.4.254: bytes=32 time<1ms TTL=255
Reply from 172.18.4.254: bytes=32 time<1ms TTL=255
Reply from 172.18.4.254: bytes=32 time<1ms TTL=255
```

PC-1 & PC-4 : "request timed out "

Il n'y a aucune différence entre le 1er test.

On met en place la table de routage du routeur-2

```
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip route 172.16.0.0 255.255.0.0 172.17.2.254
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Gateway of last resort is not set

S    172.16.0.0/16 [1/0] via 172.17.2.254
     172.17.0.0/16 is variably subnetted, 2 subnets, 2 masks
C     172.17.0.0/16 is directly connected, GigabitEthernet0/0
```

PC-1 (172.16.1.1) et sa passerelle (172.16.1.254) :

```
C:\>ping 172.16.1.254

Pinging 172.16.1.254 with 32 bytes of data:

Reply from 172.16.1.254: bytes=32 time<1ms TTL=255
Reply from 172.16.1.254: bytes=32 time<1ms TTL=255
Reply from 172.16.1.254: bytes=32 time=1ms TTL=255
Reply from 172.16.1.254: bytes=32 time<1ms TTL=255

Ping statistics for 172.16.1.254:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

PC-2 (172.17.2.2) et sa passerelle (172.17.2.254) :

```
C:\>ping 172.17.2.254

Pinging 172.17.2.254 with 32 bytes of data:

Reply from 172.17.2.254: bytes=32 time<1ms TTL=255
Reply from 172.17.2.254: bytes=32 time=1ms TTL=255
Reply from 172.17.2.254: bytes=32 time<1ms TTL=255
Reply from 172.17.2.254: bytes=32 time<1ms TTL=255

Ping statistics for 172.17.2.254:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

PC-3 (172.17.3.254) et sa passerelle (172.17.3.254) :

```
C:\>ping 172.17.3.254

Pinging 172.17.3.254 with 32 bytes of data:

Reply from 172.17.3.254: bytes=32 time<1ms TTL=255
Reply from 172.17.3.254: bytes=32 time<1ms TTL=255
Reply from 172.17.3.254: bytes=32 time<1ms TTL=255
Reply from 172.17.3.254: bytes=32 time<1ms TTL=255

Ping statistics for 172.17.3.254:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

PC-4 (172.18.4.254) et sa passerelle (172.18.4.254) :

```
C:\>ping 172.18.4.254

Pinging 172.18.4.254 with 32 bytes of data:

Reply from 172.18.4.254: bytes=32 time<1ms TTL=255
Reply from 172.18.4.254: bytes=32 time<1ms TTL=255
Reply from 172.18.4.254: bytes=32 time<1ms TTL=255
Reply from 172.18.4.254: bytes=32 time<1ms TTL=255

Ping statistics for 172.18.4.254:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Après le routage des tables des ports routeur-1 et routeur-2, on passe maintenant au dernier test de communication entre machine.

PC-1 & PC-3 :

```
C:\>ping 172.17.3.254

Pinging 172.17.3.254 with 32 bytes of data:

Reply from 172.17.3.254: bytes=32 time<1ms TTL=254
Reply from 172.17.3.254: bytes=32 time<1ms TTL=254
Reply from 172.17.3.254: bytes=32 time<1ms TTL=254
Reply from 172.17.3.254: bytes=32 time<1ms TTL=254

Ping statistics for 172.17.3.254:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

PC-2 & PC-4 :

```
C:\>ping 172.18.4.254

Pinging 172.18.4.254 with 32 bytes of data:

Reply from 172.18.4.254: bytes=32 time<1ms TTL=254
Reply from 172.18.4.254: bytes=32 time<1ms TTL=254
Reply from 172.18.4.254: bytes=32 time=8ms TTL=254
Reply from 172.18.4.254: bytes=32 time<1ms TTL=254

Ping statistics for 172.18.4.254:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 8ms, Average = 2ms
```

PC-1 & PC-4 :

```
C:\>ping 172.18.4.254

Pinging 172.18.4.254 with 32 bytes of data:

Reply from 172.18.4.254: bytes=32 time<1ms TTL=255
Reply from 172.18.4.254: bytes=32 time<1ms TTL=255
Reply from 172.18.4.254: bytes=32 time<1ms TTL=255
Reply from 172.18.4.254: bytes=32 time<1ms TTL=255

Ping statistics for 172.18.4.254:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

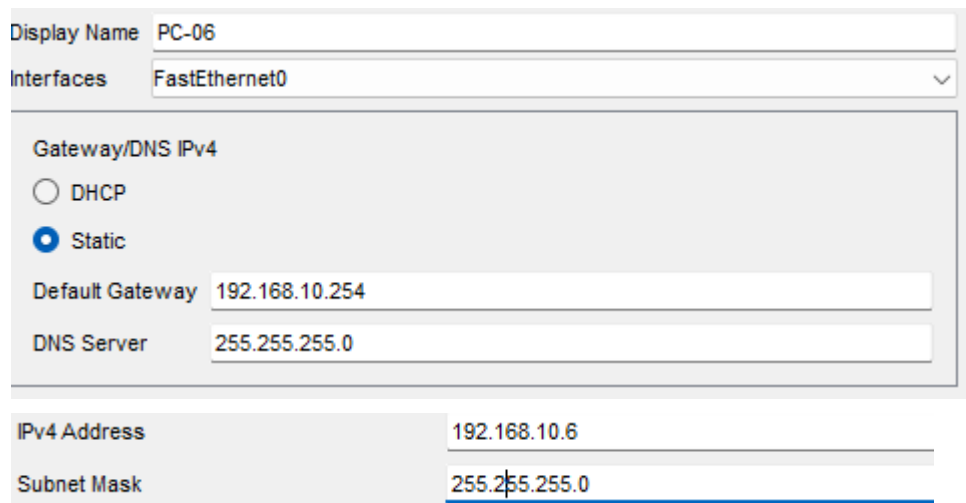
La différence que je perçois entre le routage des routeurs, les machines sont maintenant apperçu par leur adresse MAC d'un switch à un autre. "Approximate round trip times in milli-seconds:

Les prolongations

PC 05 :

Display Name	PC-5
Interfaces	FastEthernet0
Gateway/DNS IPv4	
☐ DHCP	
☒ Static	
Default Gateway	172.18.5.254
DNS Server	255.255.0.0
IPv4 Address	
172.18.5.5	
Subnet Mask	
255.255.0.0	

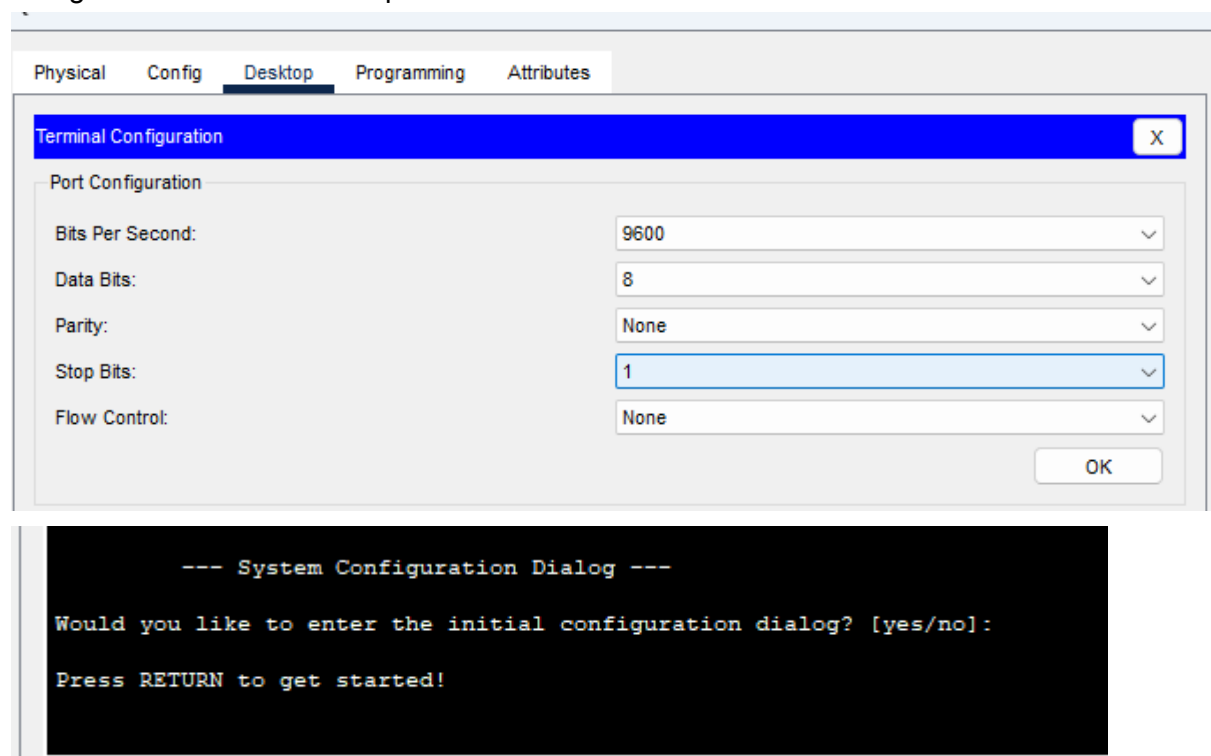
PC 06 :



The screenshot shows a configuration window for a device named 'PC-06'. The 'Interfaces' dropdown is set to 'FastEthernet0'. Under the 'Gateway/DNS IPv4' section, the 'Static' radio button is selected. The 'Default Gateway' is set to '192.168.10.254' and the 'DNS Server' is set to '255.255.255.0'. At the bottom, the 'IPv4 Address' is '192.168.10.6' and the 'Subnet Mask' is '255.255.255.0'.

Display Name	PC-06
Interfaces	FastEthernet0
Gateway/DNS IPv4	
☐ DHCP	
☒ Static	
Default Gateway	192.168.10.254
DNS Server	255.255.255.0
IPv4 Address	192.168.10.6
Subnet Mask	255.255.255.0

configuration routeur avec la passerelle PC-06 :



The top part of the image shows the 'Terminal Configuration' window of a router. The 'Desktop' tab is active. The 'Port Configuration' section shows the following settings: Bits Per Second: 9600, Data Bits: 8, Parity: None, Stop Bits: 1, and Flow Control: None. An 'OK' button is at the bottom right. The bottom part of the image shows a terminal window with the following text:

```
--- System Configuration Dialog ---  
Would you like to enter the initial configuration dialog? [yes/no]:  
Press RETURN to get started!
```

Ensuite j'ai configuré mon pc avec la méthode du câble bleu :

```

Press RETURN to get started!

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gigabitethernet 0/0
Router(config-if)#ip address
% Incomplete command.
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/1
Router(config-if)#
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/0
Router(config-if)#exit
Router(config)#interface GigabitEthernet 0/1
Router(config-if)#ip address 192.168.10.6 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

```

directement via le terminal du PC06

Je ne comprend pas pourquoi, mais je n'arrivais pas à configurer mon routeur de la même manière que le PC06, alors pour le PC05, je me dirige directement sur le terminal de command line interface du routeur 03 :

```

-----
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/1
Router(config-if)#
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/0
Router(config-if)#exit
Router(config)#interface GigabitEthernet 0/1
Router(config-if)#ip address 192.168.10.6 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

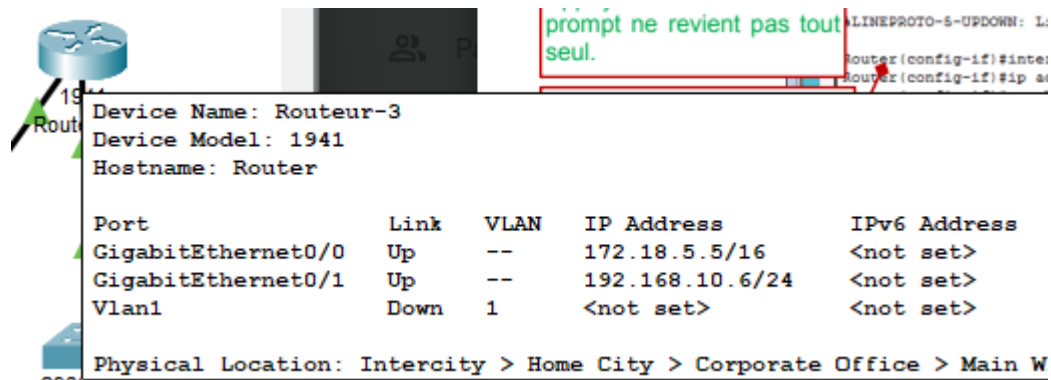
Router(config-if)#exit
Router(config)#interface GigabitEthernet 0/0
Router(config-if)#ip address 172.18.5.5 255.255.0.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

```

Tous les ports sont biens connectés :



Je modifie mes erreurs, j'ai remarqué avoir router les mauvaises adresses.

